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Problem 6 2

a)

$$\begin{aligned} E[(2 + 4X)^2] &= E[4 + 16X + 16X^2] = \\ E[4] + 16E[X] + 16E[X^2] &= \\ 4 + 16 \cdot 2 + 16 \cdot 8 &= 164 \end{aligned}$$

b)

$$\begin{aligned} E[\cancel{X}^2 + (X + 1)^2] &= E[4 + 16X + 16X^2] = \\ E[1] + 2E[X] + 2E[X^2] &= \\ 1 + 2 \cdot 2 + 2 \cdot 8 &= 21 \end{aligned}$$

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Problem 1

In order to determine the constants a and b , we use the following equations:

$$1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^1 (a + bx^2)dx = a + \frac{1}{3}b$$
$$\frac{3}{5} = \int_{-\infty}^{\infty} xf(x)dx = \int_0^1 (ax + bx^3)dx = \frac{1}{2}a + \frac{1}{4}b$$

Thus, we have a linear system of two equations with two unknowns:

$$\begin{cases} a + b/2 &= \frac{6}{5} \\ a + b/3 &= 1 \end{cases}$$

Solving the system, we obtain $a = 0.6$ and $b = 1.2$.

Problem 3

a) Find the marginal probability distributions:

Marginal probability distribution for X_1

$$P(X_1 = 0) = \frac{1}{8} + \frac{1}{16} = \frac{3}{16}$$

$$P(X_1 = 1) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$P(X_1 = 2) = \frac{3}{16} + \frac{1}{8} = \frac{5}{16}$$

$$P(X_1 = 3) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

Marginal probability distribution for X_2

$$P(X_2 = 1) = \frac{1}{8} + \frac{1}{16} + \frac{3}{16} + \frac{1}{8} = \frac{1}{2}$$

$$P(X_2 = 2) = \frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{4} = \frac{1}{2}$$

b) Find expectation, variance and covariance for X_1 and X_2

$$E[X_1] = 0 \cdot \frac{3}{16} + 1 \cdot \frac{1}{8} + 2 \cdot \frac{5}{16} + 3 \cdot \frac{3}{8} = 1.875$$

$$E[X_1^2] = 0^2 \cdot \frac{3}{16} + 1^2 \cdot \frac{1}{8} + 2^2 \cdot \frac{5}{16} + 3^2 \cdot \frac{3}{8} = 4.750$$

$$Var[X_1] = E[X_1^2] - E[X_1]^2 = 4.750 - (1.875)^2 = 1.234$$

$$E[X_2] = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 1.5$$

$$E[X_2^2] = 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{2} = 2.5$$

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$$Var[X_2] = E[X_2^2] - E[X_2]^2 = 0.25$$

$$E[X_1 X_2] = \frac{47}{16}$$

$$Cov(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2] = \frac{47}{16} - \frac{45}{16} = \frac{1}{8}$$

Problem 5

There are two ways to solve this problem. First approach: there exists a formula to calculate expectation, when we know cdf

$$E[X] = \int_0^{\infty} (1 - F(x)) dx$$

Substituting known $F(x)$ in this formula, we get:

$$E[X] = \int_0^4 \left(1 - \frac{x}{4}\right) dx = \left(x - \frac{x^2}{8}\right) \Big|_0^4 = 2$$

Second approach: we convert cdf into pdf and use the classical formula for expectation.

$$f(x) = \frac{dF(x)}{dx} = 1/4$$

when x lies in $(0, 4)$.

$$E[X] = \int_0^4 \left(x \cdot \frac{1}{4}\right) dx = \left(\frac{x^2}{8}\right) \Big|_0^4 = 2$$

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Problem ~~10~~ 4

$$E[X] = \int_0^{\infty} x e^{-x} dx = [-x e^{-x}]_0^{\infty} + \int_0^{\infty} e^{-x} dx = [-e^{-x}]_0^{\infty} = 1$$

$$E[X^2] = \int_0^{\infty} x^2 e^{-x} dx = [-x^2 e^{-x}]_0^{\infty} + 2 \int_0^{\infty} x e^{-x} dx = 2$$

$$\text{Var}[X] = E[X^2] - E[X]^2 = 2 - 1^2 = 1$$